



Human Capital Management Platform

Technical Documentation

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Executive Summary

Hire-Archy is a full-stack HC Management Platform built as a demonstration project for the technical assessment for EPI-USE Africa. It provides a platform to manage employees, displaying and changing hierarchical structures and as a bonus leave workflows.

Core Features

- Creating, reading, updating and deleting employee data.
- Interactive visual display of organisation hierarchy.
- Role-based access control. (HR, MANAGER, EMPLOYEE)
- Leave management and workflow.
- Account security and two-factor authentication.
- Soft-delete employee records with hard-delete capability.
- Gravatar integration with custom avatar and banner fallback.

User roles

Role	Access Level	Key Capabilities
HR Administrator	Full access	Create/edit/deactivate/delete employees, view all data, approve leave, full organogram
Manager	Elevated access	Edit subordinate job titles and salaries, view team organogram, approve team leave
Employee	Limited access	View own profile, view own position in organogram, apply for leave, view leave balances

Technology stack

- Next.js – App router, server components and built-in API routes.
- TypeScript – Base language.
- Tailwind CSS – Inline styling of components.
- NextAuth.js – Session authentication using JWT; OAuth provider.
- Prisma – Database ORM
- Neon PostgreSQL – Serverless database with branching for dev/prod separation.
- Vercel Blob – File storage and content delivery.
- Speakeasy – 2FA Library.
- react-organizational-chart – React library for hierarchy tree rendering.
- Vercel – Website hosting and deployment. Preview/Production environments.
- bcryptjs – Password hashing.

Database Design

The database is designed around two core entities: User for authentication and Employee for employee data. This separation follows the single responsibility principle.

Model	Purpose	Key Fields
User	Authentication & access control	email, password (bcrypt), role (HR/MANAGER/EMPLOYEE), twoFactorSecret, firstLogin
Employee	Employment data	employeeNumber, name, surname, birthDate, gross_salary, position, managerId (self-ref)
LeaveBalance	Leave entitlements	annualBalance, sickBalance, lastAccrualDate
LeaveRequest	Leave applications	leaveType, startDate, endDate, days, status (PENDING/APPROVED/REJECTED/CANCELLED)
EmployeeNumberSequence	Auto-increment tracking	lastNumber — ensures employee numbers are never reused



Key Design Decisions

Self-Referencing Reporting Line for Organogram

The employee table references itself via the managerId field, enabling an unlimited-depth org-tree as long as there are no reporting loops.

Soft Delete Protocol

Employees can be deactivated, which disables their accounts from being logged into and being edited. This preserves historical data and maintains referential integrity. If need be, a hard-delete is also available to completely remove an employee's data from the database.

Automatic Employee Number

The separate unlinked table "EmployeeNumberSequence" is a simple yet effective solution to never reuse any employee numbers even if an employee's data is hard-deleted. It counts up each time a user is created and assigns the next number.

Leave Cycle Design

The leave cycle for this fictional company is set to 2 days per month (24 days per year) from the employee's creation date. A Vercel Cron job triggers every month to accrue leave days. The use of the external date nager API allows for accurate leave applications not counting public holidays as working days.

Security

Authentication Flow

Logging into Hire-Archy uses two-factor authentication upon each login to ensure strong security.

1. Email and password verification via NextAuth.js credentials provider.
2. TOTP 2FA verification using Speakeasy library.
3. Session created with JWT containing the role and firstLogin flag which triggers if the user's password was reset and they need to re-register.

Additional Security Measures

- The proxy.ts middleware intercepts all non-API, non-static requests and forces the page to be redirected.
- All emails are normalised to lowercase before storage and lookup.
- Passwords are hashed with bcryptjs.
- Environment variables are stored in Vercel deployment.
- Separate Neon database and preview deployments in Vercel to eliminate test data contamination.

Design Patterns

Pattern	Used In	Benefit
Repository Pattern (via Prisma)	All API routes	Abstracts database queries, enables easy swapping of data sources
Server Components + Client Components	Dashboard, HR pages	Server components fetch data; client components handle interactivity
Optimistic UI Updates	Leave requests, avatar uploads	Immediate UI feedback before server confirmation improves UX
Cascade Delete	Employee hard delete	Database-level referential integrity ensures no orphaned records
Soft Delete	Employee deactivation	Preserves historical data and audit trail
JWT Session Strategy	NextAuth.js	Stateless sessions reduce database load compared to session storage
unstable_cache	Dashboard data fetching	Caches Prisma queries to reduce database round-trips
Transaction Pattern	Employee creation, leave approval	Atomic operations ensure data consistency across multiple tables

Environment & CI/CD Pipeline

The project uses a GitHub Repository for code storage and version control. Two branches exist within the repo, Main & dev. This same concept is replicated in Vercel and Neon Database to use a dev branch for testing and a main branch for production deployment.

Vercel automatically detects when a push has been made to a branch and deploys it.

Technology Justification

Web Hosting & Deployment

Vercel was chosen for web hosting and deployment over other alternatives, such as Heroku and Google Cloud, because of the rich feature set within the free hobby tier and built-in integration with Neon PostgreSQL. It centralised a lot of components and even allowed for

some extra functionality, like Vercel Blob for file storage and Vercel Cron jobs for automatic leave accruals.

Language Framework

Next.js was an easy choice because of familiarity and being a full-stack solution in a single framework. The App Router's server components eliminate the need for separate data-fetching libraries and the built-in API routes replace the need for an external server. However other frameworks were considered, such as Laravel, but I decided I want to use this opportunity to learn Next.js as well.

Database

Neon PostgreSQL was selected because of the great integration with Vercel while also being a relational model which is better for an HR solution with complex relationships. Neon's branching feature was also a key factor to allow for easier dev/prod separation.

Prisma was specifically chosen as the ORM because of the rich feature set, such as Prisma Studio. This proved to be very useful during development and local testing.

Security

TOTP was chosen as the two-factor authentication method over other methods, such as email and SMS verification because of the simplicity and security. Additional costs would have been associated with SMS verification, which is also susceptible to SIM-swapping attacks. TOTP works offline and is becoming a more common method by the day.

NextAuth.js with JWT was used because it provides a production-ready authentication solution. It handles CSRF protection, secure cookie management and session encryption. While being more complex to implement, it was deemed a must-have for a secure employee management platform that deals with sensitive information. The JWT strategy was chosen over database sessions to reduce database load. Each request doesn't require a database lookup to validate the user's role.

Conclusion

Hire-Archy demonstrates a capable and functional platform for HR representatives to manage employee data, enables managers to complete leave workflows efficiently and employees to view the organisation around them. It is production-ready with modern web technologies that prioritises security while also being easy to use.